AP Physics 1

Name _____

Period _____ Date _____

Practice Test – Introductory Dynamics (Version 3)

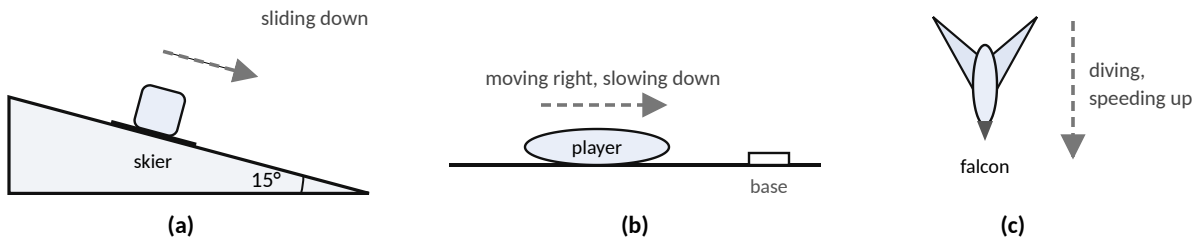
On any problems, you must show all work, include units, round properly, and box your final answer. Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

For all questions:

- Use g = 9.8 m/s² (9.8 N/kg). Air resistance is negligible unless the question says it is not.
- Free-body diagrams: draw the forces themselves, not components, each as an arrow starting on the dot.
- Round answers to the significant figures of the given values.
- 50 minutes. Four questions, 25 points each (100 points total).

1. (25 points) For each situation, draw and label the forces (not components) exerted on the object named. Draw each force as its own arrow starting on the dot and pointing away from it, with lengths that show which forces are larger.



(a) A skier slides down a 15° slope, moving down and to the right. Air resistance is negligible. Draw the free-body diagram of the skier.

(b) A baseball player slides into a base on level ground, moving to the right and slowing down. Draw the free-body diagram of the player.

(c) A falcon dives straight down and is speeding up. Air resistance is **not** negligible. Draw the free-body diagram of the falcon.

(a) skier

(b) player

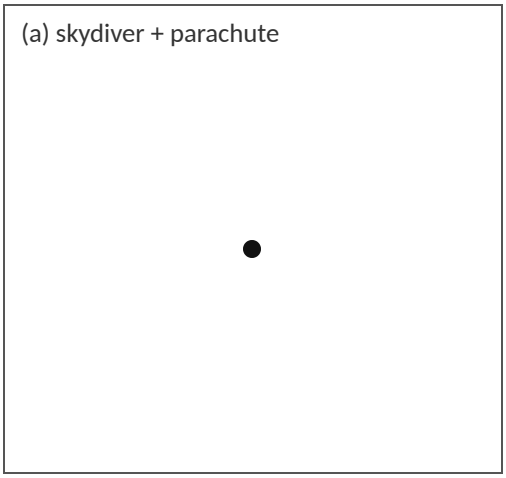
(c) falcon

(d) As the falcon in part (c) keeps diving, the drag force on it grows. What happens to the falcon's acceleration, and what happens to its speed? Explain using Newton's second law.

2. (25 points) A skydiver and parachute have a combined mass of 75.0 kg. The skydiver falls straight down and opens the parachute. Just after it opens, the drag force from the air on the skydiver and parachute is 1.50×10^3 N.



(a) Draw the free-body diagram of the skydiver and parachute just after the parachute opens. Then calculate their acceleration, magnitude and direction.

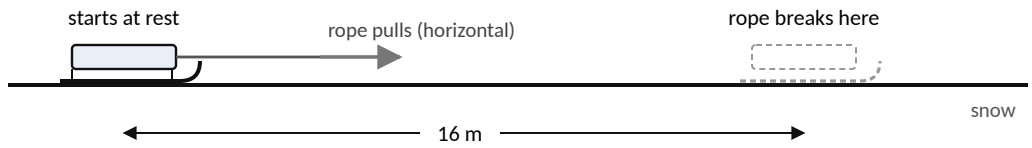


(b) Describe the motion just after the parachute opens: which way is the skydiver moving, and is the skydiver speeding up or slowing down? Explain.

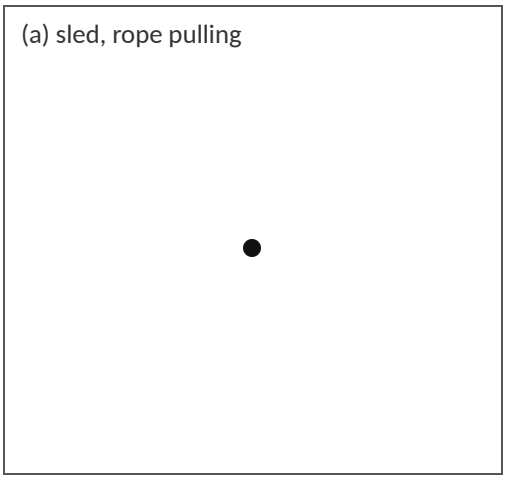
(c) Later, the skydiver falls at a constant terminal speed. Calculate the drag force at terminal speed.

(d) At the instant in part (a), the air pushes up on the skydiver and parachute. Identify the Newton's third law partner of that force: what exerts it, on what object, its magnitude, and its direction.

3. (25 points) A 24 kg sled starts from rest on level snow. A horizontal rope pulls it with a tension force of 85 N, and friction on the sled is 37 N. After the sled has moved 16 m, the rope breaks. Friction stays 37 N the whole time.



(a) Draw the free-body diagram of the sled while the rope pulls it. Then calculate the sled's acceleration.

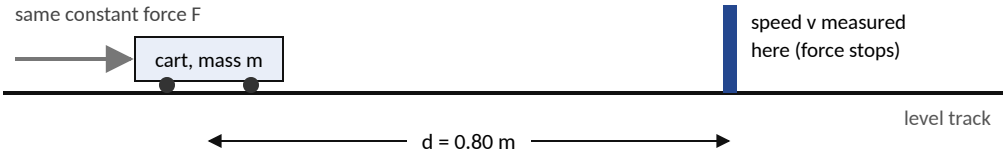


(b) Calculate the sled's speed after it has moved 16 m.

(c) The rope breaks. Calculate the sled's acceleration now, magnitude and direction.

(d) Calculate how far the sled slides after the rope breaks before it stops.

4. (25 points) Students push carts of different mass m from rest along a level track. Each cart gets the same constant force F over the same distance d = 0.80 m, and the students measure the cart's final speed v. Newton's second law and kinematics give $v^2 = 2Fd \cdot (1/m)$.

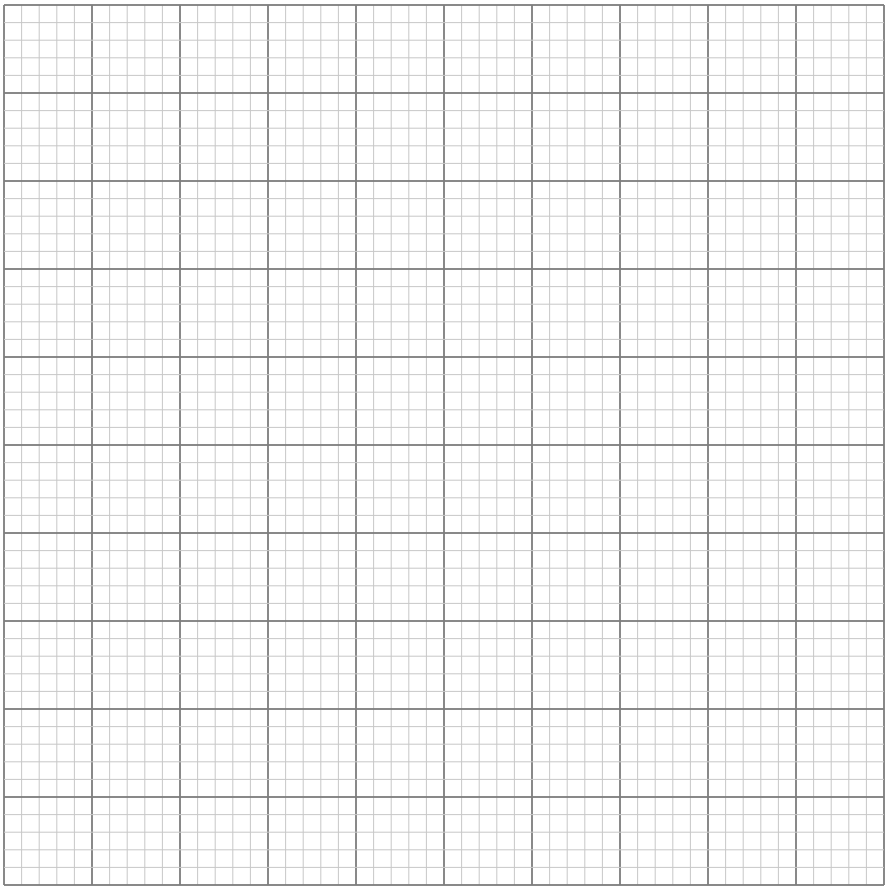


Cart mass m (kg)	Final speed v (m/s)	1/m (kg^{-1})	v^2 (m^2/s^2)
0.40	2.17		
0.60	1.82		
0.80	1.53		
1.00	1.40		
1.20	1.26		
1.40	1.19		

(a) Match $v^2 = 2Fd \cdot (1/m)$ to $y = mx + b$ so the data form a straight line. State what you will plot on the vertical axis and on the horizontal axis, and what the slope of the line represents.

(b) Calculate 1/m and v^2 for each trial and fill in the table. Show one sample calculation of each below.

(c) Plot v^2 versus 1/m on the grid. Label each axis with the quantity and units, choose a scale, and draw the best-fit line.



(d) Use two far-apart points on your line to calculate its slope, with units. Show the two points you use.

(e) Use your slope to calculate the force F that pushed the carts.

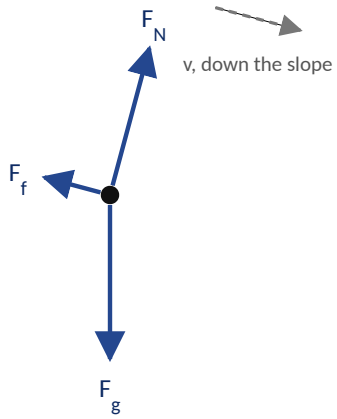
Answer Key – Practice Test, Version 3 – Introductory Dynamics

Worked step by step, the same way as the class notes: sketch, free-body diagram, $a = F_{net}/m$, forces with signs, algebra first, then numbers with units. $g = 9.8\ m/s^2$; answers match the significant figures of the givens.

Every problem: $a = \Sigma F/m$ (equation sheet), written $a = F_{net}/m$. Replace F_{net} with the actual forces and their signs ($F_D - F_g$, $F_T - F_f$, $-F_f$), swap $F_g = mg$, solve in letters, then plug in.
Kinematics, the first three on the sheet: #1 $v = v_0 + at$ · #2 $\Delta x = v_0 t + \frac{1}{2}at^2$ · #3 $v^2 = v_0^2 + 2a\Delta x$. Pick the one that leaves out the variable you don't know and don't need.

1. Free-body diagrams (25 points)

(a) skier on the 15° slope



Step 1 **List every force on the skier, each in the direction it really acts.**

Gravity F_g pulls straight down, not into the slope.

The slope pushes out, perpendicular to its surface: F_N, tilted 15° from straight up.

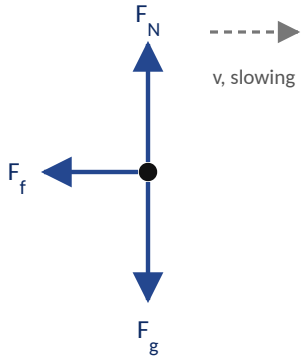
Friction from the snow F_f points up the slope, opposite the sliding.

Step 2 **Check the lengths.**

F_g is the longest arrow. F_N is a little shorter, and F_f is short. The directions are what earn the points here.

Watch out: no arrow "down the slope." Gravity already pulls the skier down the hill, and the motion itself is not a force.

(b) player, sliding right



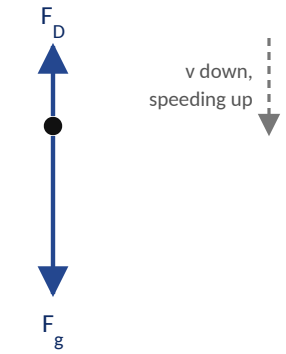
Step 3 **Vertical, then horizontal.**

No vertical acceleration, so $F_N = F_g$ (equal arrows).

Friction F_f is the only horizontal force. It points opposite the sliding: left.

Watch out: no arrow to the right. Moving right does not need a force to the right; the player keeps sliding because of inertia.

(c) falcon, diving



Step 4 **Two forces: gravity and drag.**

Gravity F_g pulls down. Air resistance F_D pushes up, opposite the motion.

Speeding up while moving down means the acceleration points down. $a = F_{net}/m$, so F_{net} points down too: $F_g > F_D$. The F_g arrow is longer.

Step 5 **(d) Drag grows: follow $a = F_{net}/m$. Down is +.**

$a = \frac{F_{net}}{m} \rightarrow a = \frac{F_g - F_D}{m}$. F_g stays the same while F_D grows, so $F_g - F_D$ shrinks.

The acceleration decreases toward zero (it still points down).

The acceleration still points the same way as the velocity, so the falcon keeps speeding up, just more slowly. When $F_D = F_g$, $a = 0$.

The speed increases until it reaches terminal speed, then stays constant

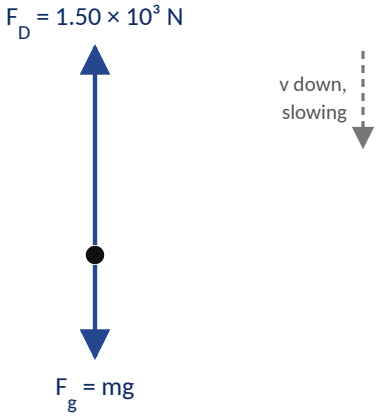
Watch out: a smaller acceleration does not mean slowing down. An object slows down only when its acceleration points opposite its velocity.

2. Parachute opens (25 points)

Step 1 **The object is the skydiver and parachute together. Up is +.**

known	wanted
$m = 75.0$ kg, $F_D = 1.50 \times 10^3$ N up (just after opening), $F_g = mg$ down	a, then the motion, then F_D at terminal speed

(a) just after opening



F_D is about twice F_g, so the up arrow is about twice as long. The net force is up.

Step 2 **(a) Newton's second law, letters first.**

$$a = \frac{F_{net}}{m} \rightarrow a = \frac{F_D - F_g}{m} = \frac{F_D - mg}{m}$$

$$a = \frac{1.50 \times 10^3 \text{ N} - (75.0 \text{ kg})(9.8 \text{ m/s}^2)}{75.0 \text{ kg}} = \frac{1500 \text{ N} - 735 \text{ N}}{75.0 \text{ kg}} = \frac{765 \text{ N}}{75.0 \text{ kg}} = 10.2 \text{ m/s}^2$$

$a = 10.2$ m/s², upward

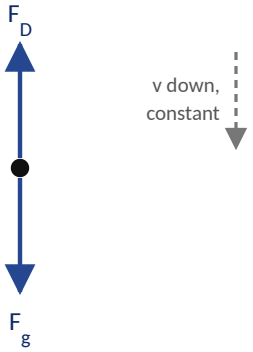
Step 3 **(b) Velocity and acceleration point opposite ways.**

The skydiver was falling, so the velocity still points down. The acceleration points up, opposite the velocity.

Still moving down, and slowing down

Watch out: an upward acceleration does not mean the skydiver moves up. It means the downward speed is getting smaller.

(c) at terminal speed



Step 4 **(c) Terminal speed: constant velocity, so a = 0.**

$a = \frac{F_{net}}{m} \rightarrow 0 = \frac{F_D - mg}{m} \rightarrow F_D - mg = 0 \rightarrow F_D = mg$

$F_D = (75.0\text{ kg})(9.8\text{ m/s}^2) =$ **735 N, upward**

Equal arrows: the drag has dropped from 1.50×10^3 N to the weight as the skydiver slowed down.

Step 5 **(d) Third law last: same size, opposite direction, on the other object.**

$F_{A\text{ on }B} = -F_{B\text{ on }A}$. The partner of $F_{\text{air on skydiver}}$ is $F_{\text{skydiver on air}}$.

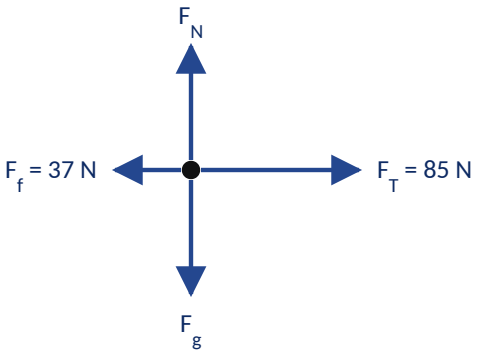
The skydiver and parachute push down on the air, 1.50×10^3 N, downward

Watch out: the partner of drag is not gravity. Gravity is the Earth pulling on the skydiver; its partner is the skydiver pulling up on the Earth.

3. Sled and rope (25 points)

Step 1 **The + direction is the way the sled moves (right).**

(a) sled, rope pulling



F_N and F_g are equal (no vertical acceleration). F_T to the right is longer than F_f to the left, so the net force is to the right.

Step 2 **(a) Newton's second law, horizontal.**

$a = \frac{F_{net}}{m} \rightarrow a = \frac{F_T - F_f}{m} = \frac{85\text{ N} - 37\text{ N}}{24\text{ kg}} = \frac{48\text{ N}}{24\text{ kg}} =$ **2.0 m/s^2** (forward)

Step 3 **(b) Time is not given and not needed, so use #3.**

v_0	v	a	t	Δx
0 (rest)	?	+2.0 m/s²	not given	+16 m

$v^2 = v_0^2 + 2a\Delta x \rightarrow v^2 = 0 + 2a\Delta x \rightarrow v = \sqrt{2a\Delta x}$

$v = \sqrt{2(2.0\text{ m/s}^2)(16\text{ m})} = \sqrt{64\text{ m}^2/\text{s}^2} =$ **8.0 m/s**

Step 4 **(c) The rope breaks: friction is the only horizontal force.**

$a = \frac{F_{net}}{m} \rightarrow a = \frac{-F_f}{m} = \frac{-37\ \text{N}}{24\ \text{kg}} = -1.54\ \text{m/s}^2 \rightarrow$ **$a = -1.5\ \text{m/s}^2$** (1.5 m/s², backward, opposite the motion)

Keep −1.54 m/s² for the next step; round only at the end.

Watch out: the sled does not stop at once when the rope breaks. It keeps moving forward and slows down because of friction.

Step 5 **(d) New stage: starts at 8.0 m/s, ends at rest. Time is not given, so use #3.**

v_0	v	a	t	Δx
+8.0 m/s	0 (stops)	−1.54 m/s²	not given	?

$v^2 = v_0^2 + 2a\Delta x \rightarrow 0 = v_0^2 + 2a\Delta x \rightarrow \Delta x = \frac{-v_0^2}{2a}$

$\Delta x = \frac{-(8.0\ \text{m/s})^2}{2(-1.54\ \text{m/s}^2)} = \frac{-64\ \text{m}^2/\text{s}^2}{-3.08\ \text{m/s}^2} = 20.8\ \text{m} \rightarrow$ **$\Delta x = 21\ \text{m}$**

Two significant figures, matching 24 kg, 85 N, 37 N and 16 m.

Watch out: the 2.0 m/s² is only for while the rope pulls. After it breaks, use the new acceleration.

4. Linearize the data (25 points)

Step 1 **(a) Match $v^2 = 2Fd \cdot (1/m)$ to $y = mx + b$.**

$v^2 = 2Fd \cdot \frac{1}{m} + 0$. (It comes from $a = F/m$ and #3 with $v_0 = 0$: $v^2 = 2(F/m)d$.)

vertical (y)	horizontal (x)	slope	intercept
v^2 (m^2/s^2)	$1/m$ (kg^{-1})	$2Fd$	0

Plot v^2 on the vertical axis and $1/m$ on the horizontal axis; the slope is $2Fd$

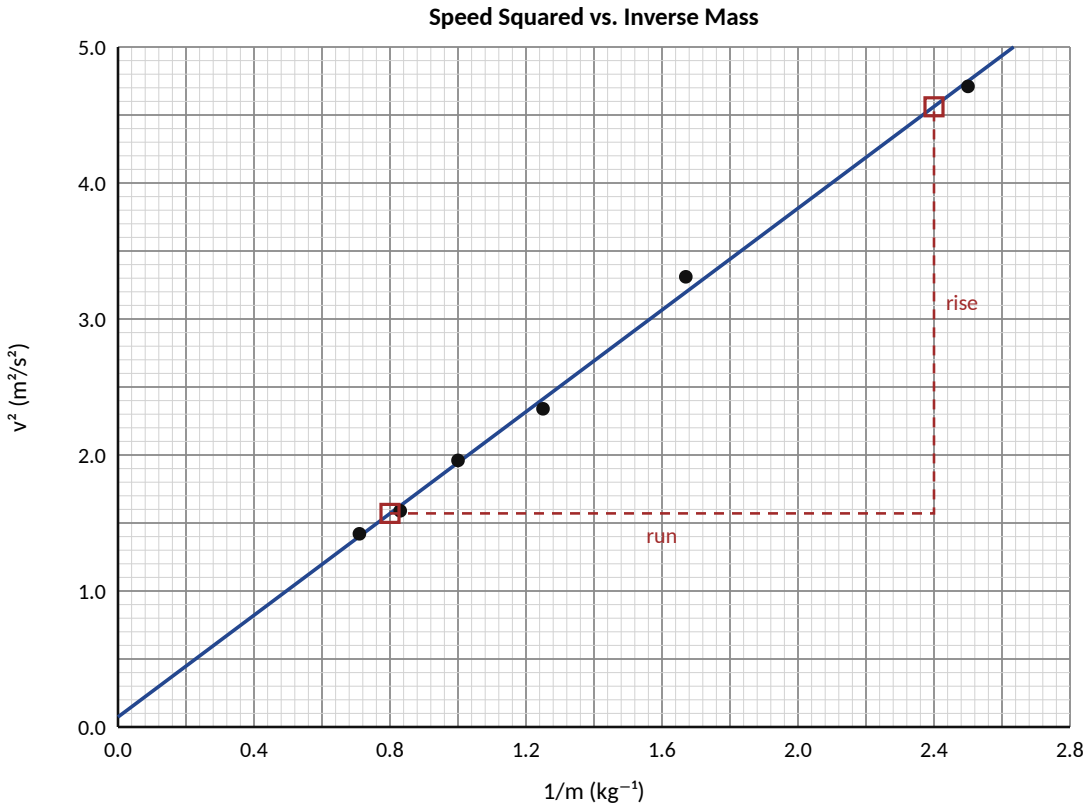
Watch out: v vs. m and v^2 vs. m are both curves, so neither is the line.

Step 2 **(b) Fill in both columns.**

Sample: $1/m = 1/(0.40\ \text{kg}) = 2.50\ \text{kg}^{-1}$ and $v^2 = (2.17\ \text{m/s})^2 = 4.71\ \text{m}^2/\text{s}^2$

m (kg)	v (m/s)	1/m (kg^{-1})	v^2 (m^2/s^2)
0.40	2.17	2.50	4.71
0.60	1.82	1.67	3.31
0.80	1.53	1.25	2.34
1.00	1.40	1.00	1.96
1.20	1.26	0.83	1.59
1.40	1.19	0.71	1.42

Step 3 **(c) Graph: labeled axes with units, a scale that uses more than half the grid, one straight line through the middle of the points.**



The line is not forced through the origin. The red squares are points read off the line, not data points.

Step 4 **(d) Slope from two far-apart points on the line, units carried.**

Points on the line: (0.80 kg^{-1}, 1.57 m^2/s^2) and (2.40 kg^{-1}, 4.56 m^2/s^2)

$$\text{slope} = \frac{4.56\ m^2/s^2 - 1.57\ m^2/s^2}{2.40\ kg^{-1} - 0.80\ kg^{-1}} = \frac{2.99\ m^2/s^2}{1.60\ kg^{-1}} = 1.869\ kg\cdot m^2/s^2$$

Units: $(m^2/s^2) \div (1/kg) = kg\cdot m^2/s^2 = (kg\cdot m/s^2)\cdot m = N\cdot m$. **slope ≈ 1.87 N·m**

Any value from 1.72 N·m to 2.02 N·m from a reasonable line earns credit.

Step 5 **(e) The slope is 2Fd. Solve for F in letters, then plug in.**

$$\text{slope} = 2Fd \rightarrow F = \frac{\text{slope}}{2d} = \frac{1.87\ N\cdot m}{2(0.80\ m)} = \frac{1.87\ N\cdot m}{1.60\ m} = 1.169\ N \rightarrow \mathbf{F \approx 1.2\ N}$$

Any value from 1.1 N to 1.3 N that follows from the student's own slope earns credit.

Watch out: the slope is 2Fd, not F. Divide by 2d = 1.60 m, not by 0.80 m.

1. Free-body diagrams

Suggested partial credit	pts
(a) F_g straight down and F_N perpendicular to the slope (not straight up)	4
(a) F_f up the slope, opposite the motion	3
(a) No components and no extra "down the slope" force	2
(b) F_N up and F_g down, equal lengths	3
(b) Friction to the left, and no force to the right	3
(c) F_g down and F_D up, each starting on the dot, labeled	3
(c) F_g longer than F_D (speeding up downward)	2
(d) Acceleration decreases toward zero, because $F_g - F_D$ shrinks	3
(d) Speed keeps increasing until terminal speed, then stays constant	2
Total	**25**

2. Parachute opens

Suggested partial credit	pts
(a) F_D up, about twice as long as F_g down; labeled	4
(a) $a = (F_D - mg)/m$ in letters first	4
(a) Substitution with units in every parenthesis	2
(a) 10.2 m/s², upward	3
(b) Still moving down	2
(b) Slowing down, because the acceleration is opposite the velocity	3
(c) $a = 0$ at terminal speed, so $F_D = mg$	3
(c) 735 N	2
(d) Skydiver and parachute on the air, 1.50×10^3 N, downward	2
Total	**25**

3. Sled and rope

Suggested partial credit	pts
(a) F_T right longer than F_f left; F_N and F_g equal; labeled	4
(a) $a = (F_T - F_f)/m$ in letters first	3
(a) 2.0 m/s²	2
(b) Picks #3 with $v_0 = 0$: $v = \sqrt{(2a\Delta x)}$	3
(b) 8.0 m/s	2
(c) $a = -F_f/m$, friction the only horizontal force	3
(c) 1.5 m/s², backward (opposite the motion)	2
(d) #3 with $v_0 = 8.0$ m/s and $v = 0$, solved for Δx in letters	3
(d) 21 m	3
Total	**25**

4. Linearize the data

Suggested partial credit	pts
(a) v^2 on the vertical axis, 1/m on the horizontal axis	3
(a) Slope = 2Fd	3
(b) 1/m column correct	2
(b) v^2 column correct	2
(c) Axes labeled with quantity and units	2
(c) Scale uses more than half the grid	2
(c) Points plotted correctly	2
(c) One straight best-fit line, not forced through the origin	2
(d) Two far-apart points on the line; slope 1.72 N·m to 2.02 N·m with units	3
(e) F = slope/(2d) in letters first	2
(e) F between 1.1 N and 1.3 N	2
Total	**25**